

Massachusetts General Claim Cost

*Static dataset pulled from Maven Analytics

Dataset Source:

<https://mavenanalytics.io/data-playground/hospital-patient-records>

About the Data Source:

<https://synthea.mitre.org/downloads>

- Data hosted within SyntheticMass has been generated by ([SyntheaTM](#)), an open-source patient population simulation made available by ([The MITRE Corporation](#)).
- The data is free from cost, privacy, and security restrictions. It can be used without restriction for a variety of secondary uses in academia, research, industry, and government.
- Jason Walonoski, Mark Kramer, Joseph Nichols, Andre Quina, Chris Moesel, Dylan Hall, Carlton Duffett, Kudakwashe Dube, Thomas Gallagher, Scott McLachlan, Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record, *Journal of the American Medical Informatics Association*, Volume 25, Issue 3, March 2018, Pages 230–238, <https://doi.org/10.1093/jamia/ocx079>

About the Data:

- Synthetic data on ~1k patients of Massachusetts General Hospital from 2011-2022, including information on patient demographics, insurance coverage, and medical encounters and procedures.

Patients

1,135

Unique patients

Encounters

27,891

Total Encounters

Procedures

47,701

Total procedures

Date Range

2011 - 2022

Key Definitions (as reported in the data dictionary csv provided with the project)

- Payer - related to the entity that is assisting the patient with coverage
- Payer Coverage - amount of cost covered by the payer
- Procedure - procedure data included in surgeries
- Encounter Class - class of the encounter, such as ambulatory, emergency, inpatient, wellness, or urgent care
- Total Claim Cost - total cost of the encounter, including all line items
- Coverage Gap visualization - shows total costs paid by patients vs. payers, segmented by payer

Details on the Readmissions data:

- Readmissions by definition means a patient was discharged from a hospital stay ad returned within 30 days
 - For this reason only the "inpatient" encounter was counted

Massachusetts General Claim Cost Analysis

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Total Claim Costs

\$101.51M

All encounters

Average Claim Cost

\$3,640

Per visit

Payer Coverage Rate

31%

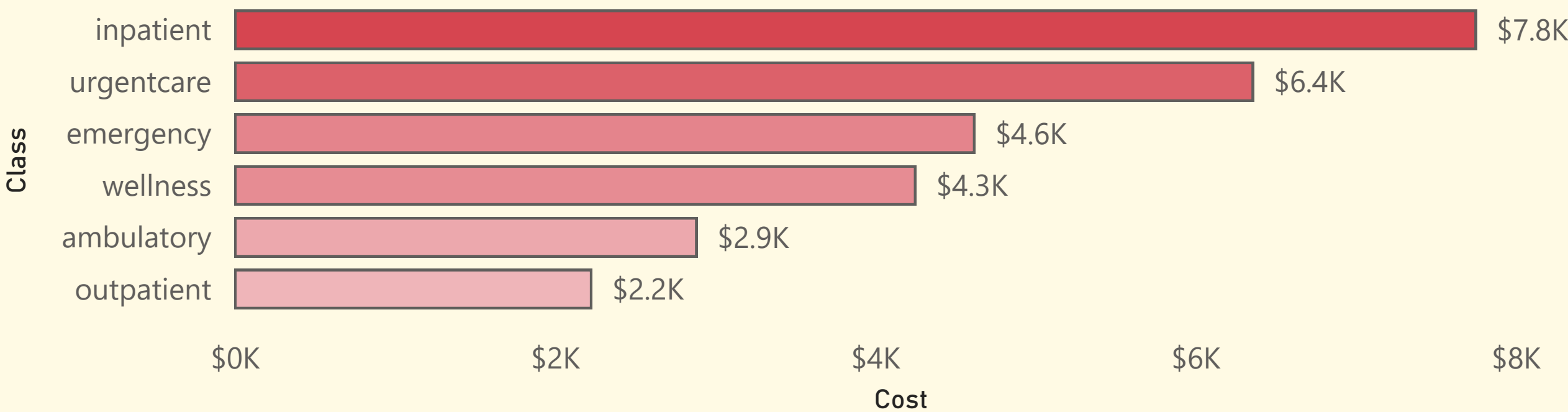
Responsible for payer (insurance)

Out-Of-Pocket Burden

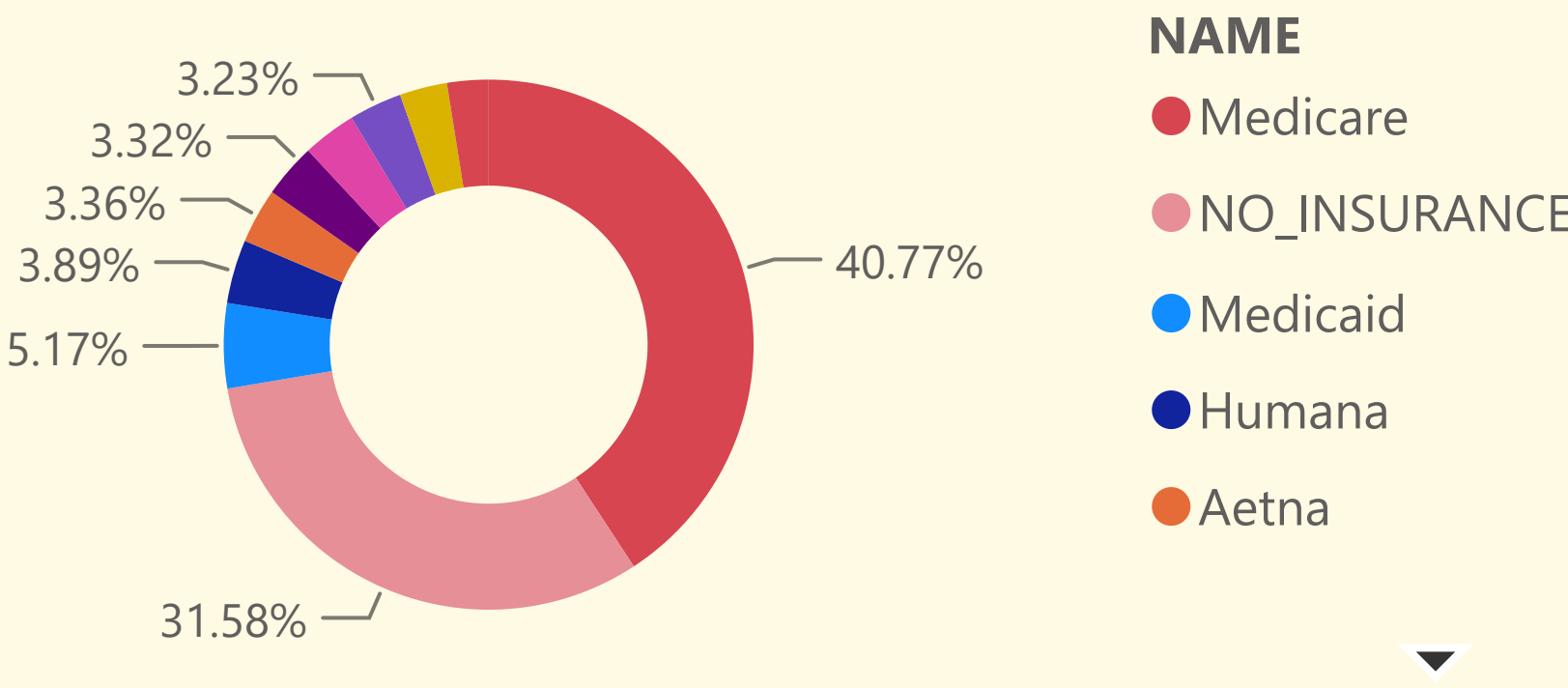
70.42M

Total

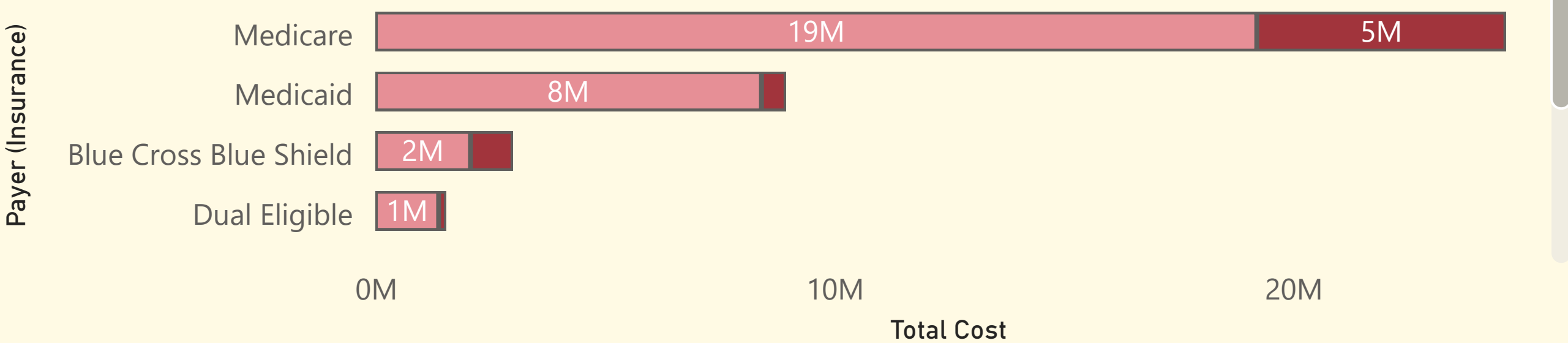
Average Claim Cost by Encounter Class



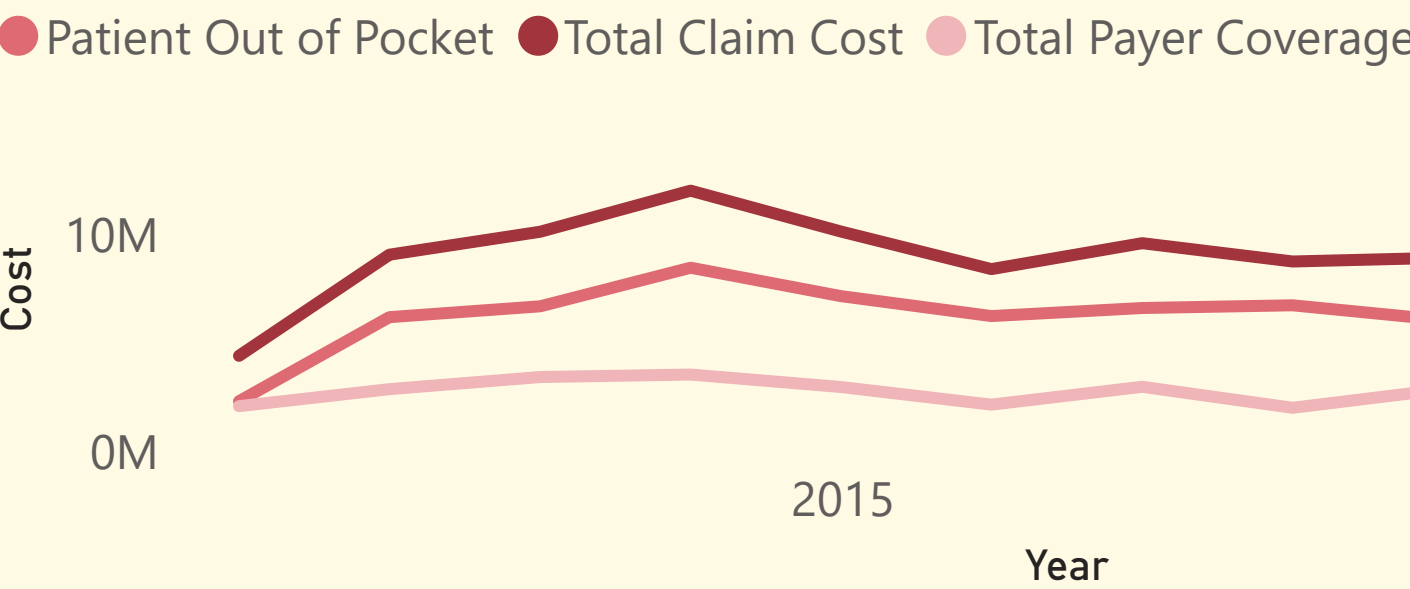
Payer Market Share by Organization



Coverage Gap by Payer



Cost Trends Over Time (2011-2022)



Massachusetts General Procedure Coverage Statistics

*Static dataset pulled from Maven Analytics

Total Procedures

47,701

Covered Procedures

24,791

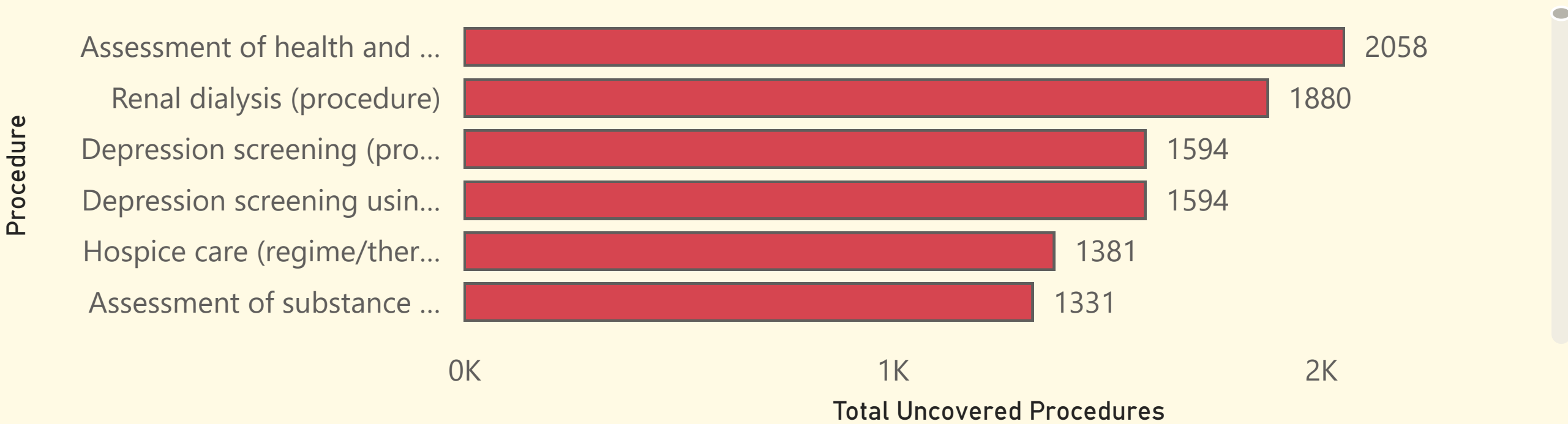
Uncovered Procedures

22,910

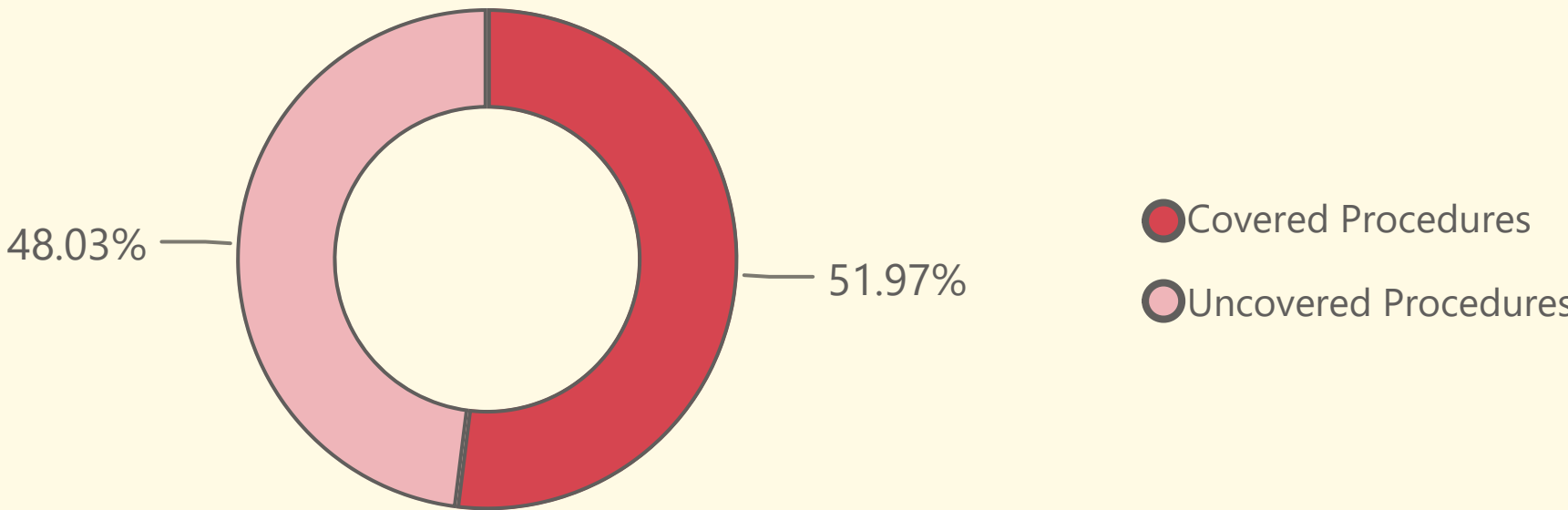
Average Claim Cost

\$3,640

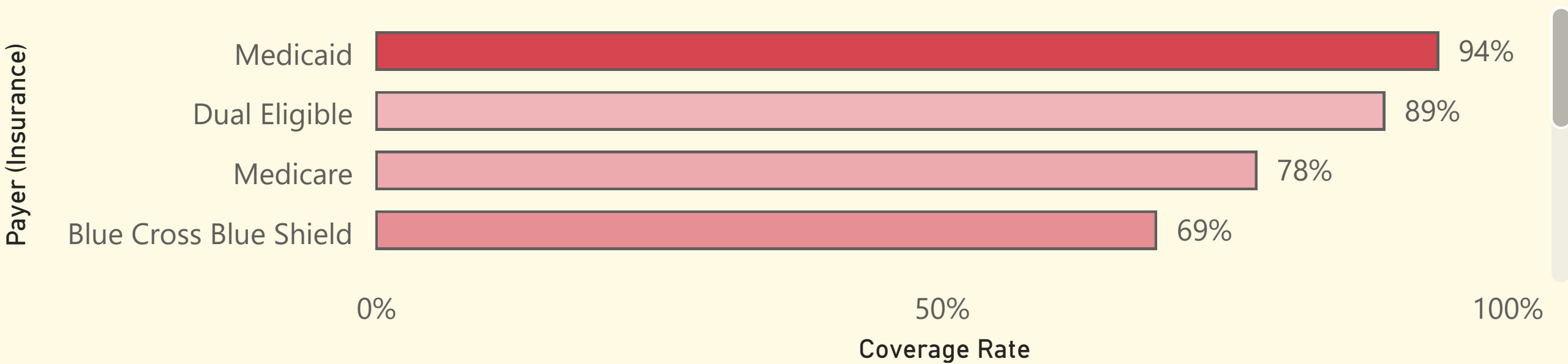
Uncoverd Procedures by Volume



Payer Market Share by Organization



Coverage Rate by Payer (Insurance)



Average Covered Cost

\$1,643

Average Uncovered Cost

\$2,828

Massachusetts General Readmission Statistics

*Static dataset pulled from Maven Analytics

Total Admissions

1135

Total Readmissions

523

Readmission Rate

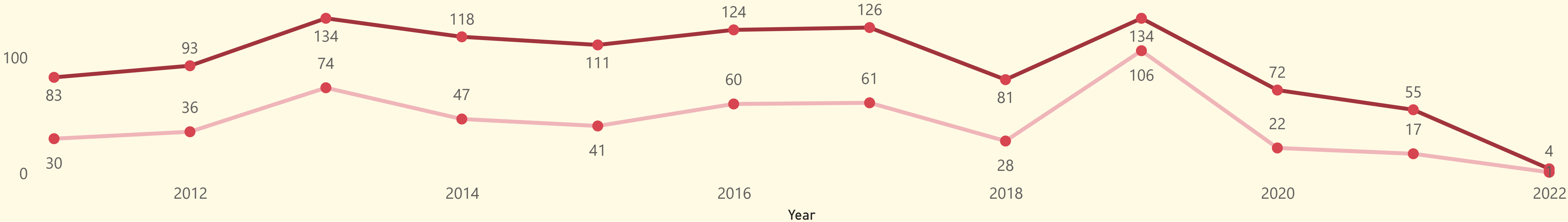
46%

Average Length of Stay

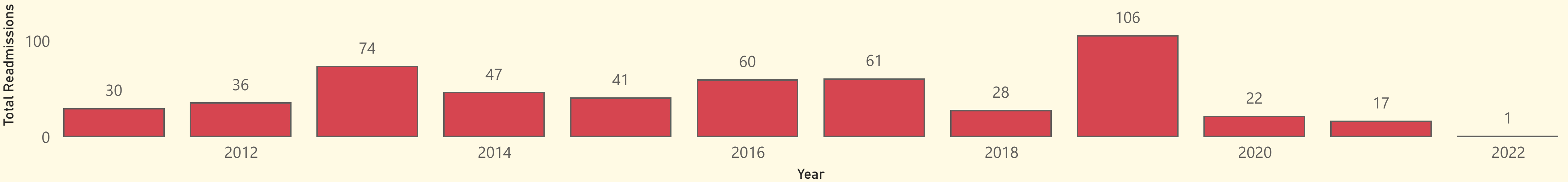
0.30

Days

Admission vs Readmissions Over Time



Readmissions by Year



Case Study Questions

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Q: How many patients have been admitted and readmitted over time?

A: total inpatient admissions reached 1135 with 523 readmissions (46% readmission rate)

Q: How long are patients staying in the hospital on average?

A: 0.3 days

Q: How much is the average cost per visit?

A: \$3,640 per visit

Q: How many procedures were covered by insurance?

A: 24,791 (51.9%) procedures were covered by insurance

Data Visualization Notes

- Analysis was scoped to answer the specific business questions outlined in the case study
- Metrics, visuals, and calculations were selected based on their relevance to the stated objectives
- Additional exploratory analysis was not performed outside the scope of the case study requirements
- Patient demographic variables (such as gender, or location) were not analyzed in depth, as they were not necessary to address the provided questions.